

February 24, 2022

Dr. Stephanie Johnson
U.S. Department of Energy
Office of Energy Efficiency and Renewable Energy
Building Technologies Office EE-2J
1000 Independence Avenue, SW
Washington, DC 20585-0121

Submitted via www.regulations.gov: Docket ID EERE-2017-BT-TP-0018

Re: Madison IAQ Comments in Response to the Request for Information for the Test Procedure and Energy Conservation Standard for Consumer Air Cleaners [*Docket ID EERE-2021-BT-STD-0035 and EERE-2021-BT-TP-0036*]

Dear Dr. Johnson:

Madison Indoor Air Quality (MIAQ) respectfully submits these comments in response to the U.S. Department of Energy's (DOE) *2022-1-26 Energy Conservation Program: Test Procedure for Direct Expansion-Dedicated Outdoor Air Systems; Supplemental Notice of Proposed Rulemaking and Request for Comment* as posted in the *Federal Register* on December 23, 2021.

MIAQ is one of the largest and most successful privately held companies in the world with a significant footprint in the HVAC market. MIAQ's mission is to make the world safer, healthier, and more productive by creating innovative solutions that deliver outstanding customer value. MIAQ's portfolio comprises of at least 15 companies including Big Ass Fans, Broan NuTone, UV Resources, and Therma-Stor. Through these companies, MIAQ offers products that could be impacted by changes to DOE's Energy Conservation Standard for consumer air cleaners.

Due to our continuous review of market demands and advancement to higher efficiency equipment, MIAQ feels uniquely qualified to provide feedback to DOE staff on the issues they wish to address. We thank DOE staff for taking time to read these comments and encourages them to work with Madison IAQ to address our thoughts and concerns during the rule-making process.

Summary

MIAQ strives to deliver safe, healthy air to customers of space heating and space cooling technologies. While DOE is chiefly concerned with defining and covering "air cleaners," the potential exists for air filters, and other equipment, to be improperly included in this regulation. MIAQ's response to the NOPD sought to draw the distinction between portable air cleaners, which have fan and air cleaning components, and in-duct products, with only air cleaning components without an energy-consuming fan. Some in-duct products utilize electrical arrays, UV lights, and Ion generators. While we have strived to provide meaningful responses to these RFIs, depending on the scope set, this rulemaking has the potential

to significantly impact, sideswipe, or possibly to not impact MIAQ companies at all. Given this potential spectrum, responding to this RFI is extremely difficult for without a defined scope and finalized determination. MIAQ recommends DOE issue a revised RFI if DOE determines that coverage is warranted beyond portable products before proceeding with its typical rulemaking process for both test procedures and standards.

MIAQ offers the following general comments on the RFI. Below that, MIAQ provides responses to specific items raised by DOE and identified, as applicable, by the specific item numbers.

MIAQ Concerns

DOE Should Follow the Rulemaking Process It Recently Set To Obtain Meaningful Feedback

DOE has abandoned its recently revised process rule, justifying this decision, “opting to deviate from this step because DOE believes that providing an opportunity for comment on potential test procedure and energy conservation standards prior to a final coverage determination for consumer air cleaners allows stakeholders an earlier opportunity to provide comment, information, and data that may help inform DOE's priority setting.”¹ Unresolved issues regarding scope and applicability from the notice of proposed determination make it difficult for stakeholders, particularly small businesses, to participate meaningfully in commenting and to provide the types of substantive technical comments necessary to determine whether a particular test procedure is feasible and the impact of energy conservation on their products. It is nearly impossible for stakeholders to provide meaningful feedback to ensure new energy conservation standards and test procedures meet applicable statutory criteria in EPCA (e.g., standards result in the maximum improvement in energy efficiency that is technologically feasible and economically justified) with so many open questions on the scope. An energy conservation standard for portable and non-portable, or incidental air cleaners, would be very different and not appropriately included in the same regulation. Unfortunately, without decisions on scope, MIAQ's feedback to the DOE's many questions will be limited.

MIAQ is dismayed by the shortening of the 75-day comment period for pre-NOPR rulemaking documents for standards to 30-days and also by the combination with the test procedure RFI. The breadth of information DOE seeks to collect is astounding for products outside of the U.S. Environmental Protection Agency's (EPA) ENERGY STAR® Program (ENERGY STAR Program). For example, this shortened comment period did not allow MIAQ to investigate test lab capacity or capabilities. Many of the manufacturers that may be scoped in, are completely unfamiliar with portable unit test procedures. While MIAQ declines to opine if classifying *portable* air cleaners as a covered product is necessary or appropriate to carry out the purposes of EPCA, there exists an appropriate test procedure,² the Association of Home Appliance Manufacturers' (“AHAM's”) *Directory of Certified Portable Electric Room Air Cleaners*, and an ENERGY STAR Program for these products. Finalizing coverage on this subset of equipment and narrowing the scope to define these products seems like the logical first step of regulation. Section 5(c) of 10 CFR Part 430, Subpart C, Appendix A provides DOE options for expanding the scope of coverage if it is deemed necessary and appropriate. At this time, however, it is neither necessary nor appropriate to include ducted air cleaners, or any of the other incidental air cleaning products DOE has scoped in with the broad

¹ 87 FR 3702, 3706

² ANSI/AHAM AC-1-2020 *Portable Household Electric Room Air Cleaners* establishes a uniform, repeatable procedure or standard method for measuring specified product characteristics of household portable air cleaners.

proposed determination. For these non-portable products, there is no test procedure used in the US that would be comparable to the established portable test procedure.

DOE must revise its definition to exclude certain products

DOE's analysis of the air cleaner market, and its ability to analyze the market, relies on the ENERGY STAR program for portable products. MIAQ does not dispute the market or energy savings potential for portable products within the scope of the ENERGY STAR program; however, the proposed definition of "air cleaner" would include other products that are not appropriately tested by the Association of Home Appliance Manufacturers (AHAM) test procedure or covered by the ENERGY STAR program. In the NOPD, DOE explains that "ENERGY STAR describes room air cleaners as portable, electric appliances that remove fine particles, such as dust and pollen, from indoor air."³ The current ENERGY STAR Product Specification defines "room air cleaner" as "an electric appliance with the function of removing particulate matter from the air and which can be moved from room to room," consistent with ANSI/AHAM AC-1-2020.⁴

MIAQ disagrees with DOE's proposed inclusion of "non-portable" air cleaners, such as those that are mounted on walls and ceilings, or that provide whole-home air cleaning in conjunction with central heating or air conditioning systems. Many of these products are not appropriate to include in this proposed air cleaners determination. The proposed expansion of the established scope to include portable products would include products DOE does not have the time or resources to quickly or easily include in the robust analysis necessary to justify their inclusion.

Further, the AHAM test method suitable for portable products does not adequately cover non-portable products, and products over 450 cubic feet per minute (CFM) are not covered by the ENERGY STAR program. The construct of the AHAM test, effectively testing the energy required to clean the air in a room, is not appropriate for larger products or very small products as the room size is fixed. There is also a difference in the particles targeted for testing when comparing the AHRI or ASHRAE procedures to the AHAM procedure. In addition, the AHAM test is not appropriate for other products potentially included in the broad proposed definition such as oven hoods, or fresh air ventilators. Such products would need to have a unique test method developed – further reason to narrow the scope and exclude products which really aren't considered air cleaners by consumers.

Commercial products should be clearly excluded. Air cleaners are not on the statutory list of commercial/industrial equipment permissible for regulation.⁵ The determination of energy savings would be substantially different for commercial products, an arduous process DOE has not made public if it has even begun the analysis. Consumer and commercial products are fundamentally different and have different statutory requirements. The Energy Policy and Conservation Act (EPCA) explicitly identifies and

³ U.S. DOE, Proposed Determination of Air Cleaners as a Covered Consumer Product, 86 Fed. Reg. 51629, 51632 (Sept. 16, 2021).

⁴ See ENERGY STAR Program Requirements for Room Air Cleaners, Eligibility Criteria Version 2.0 (Rev. April 2021), available at www.energystar.gov/sites/default/files/ENERGY%20STAR%20Version%202.0%20Room%20Air%20Cleaners%20Specification_Rev%20April%202021_with%20Partner%20Commitments.pdf.

⁵ Permissible covered products can only include one of the following types of industrial equipment: compressors; fans; blowers; refrigeration equipment; electric lights and lighting power supply circuits; electrolytic equipment; electric arc equipment; steam boilers; ovens; kilns; evaporators; dryers; and other motors. (42 U.S.C. § 6311(2)(B)). Air cleaners are not on this list.

defines certain products as covered products and provides specific statutory definitions. It also grants the Secretary the authority to add other consumer products or commercial equipment to this list of covered products, if they meet the statutory definition of consumer product, and the Secretary makes the required determinations under 42 U.S.C. § 6292(b). That is the authority that DOE has to determine whether products are consumer products, and if it makes that determination, it may add those products to the list of covered products if the requisite showings of section 6292(b) are made. Commercial products typically cannot be purchased by consumers, for consumer applications. Commercial air purifiers are meant for installation, typically in hospitals, airports, commercial buildings, and laboratories. These units have a high airflow leading to a capacity not meant for a residential use. MIAQ considers those high airflow products as industrial/commercial grade purifiers, not suitable for residential applications. If a 500 CFM air purifier were to be installed in a typical 1000 ft² home with 12 ft high ceilings it would provide 2.5 air changes per hour (ACH) and nearly seven times the ventilation required.⁶ Consumers are unable to purchase these directly for personal use and installation in their homes. Commercial units are sold through a business-to-business sales channel. DOE has not conducted the appropriate analysis to include commercial air cleaners, nor are existing test procedures adequate for this equipment.

MIAQ's proposed revisions to the definition DOE proposed in the NOPD are below, shown in ~~strikeout~~/underline:

Air cleaner is defined as a consumer product that:

- (1) Is a self-contained, mechanically encased assembly;
- (2) Is directly powered by 120 V, single-phase electric current supplied by a NEMA 1–15P or 5–15P plug;⁷
- (3) Is designed to supply airflow less than or equal 400 CFM;
- ~~(34)~~ Removes, destroys, or deactivates particulates and microorganisms from the air;
- (5) Is marketed for the primary functions of removing, destroying, or deactivating particulates and microorganisms from the air in a residential occupancy
- ~~(46)~~ Excludes products that destroy or deactivate particulates and microorganisms solely by means of ultraviolet light without a fan for air circulation; and
- (7) Excludes products that destroy or deactivate particulates and microorganisms solely by means of ultraviolet light or electrostatic air filters with or without a fan for air circulation;
- (8) Excludes products without a fan;
- (9) Excludes Incidental Air Cleaning Products; and
- ~~(510)~~ Excludes central air conditioners, room air conditioners, portable air conditioners, dehumidifiers, and furnaces as defined in 10 CFR 430.2.

Incidental Air Cleaning Product is defined as a consumer product that would meet the definition of an Air Cleaner, but which provides an additional function, not related to air purification, within the same housing, such as a vacuum cleaner, fresh air ventilators, oven hood, refrigerator, or desiccant dehumidifier, and whose air purification function is incidental to its other functions.

⁶ ANSI/ASHRAE Standard 62.2-2016, Ventilation for Acceptable Indoor Air Quality in Low-Rise Residential Buildings is the recognized standard for ventilation system design and acceptable indoor air quality (IAQ) in residential occupancies. Table 4-1a provides ventilation air requirements by floor area and number of bedrooms. For a 1,000 ft² home, the ventilation air requirements are between 45 and 75 CFM, depending on the number of bedrooms.

⁷ The “plug-in” designation is “NEMA 1–15P or 5–15P plug” per the DOE Final Rule for Uninterruptible Power Supplies, 85 FR 1447 (January 10, 2020)

The proposed modifications to seek to make clear that the scope would apply to products DOE can justify including – chiefly portable, plug-in, products – that are ripe for regulation. Specifying the voltage would appropriately exclude 240V products or damper/filter combination products which would become subject to the regulation if they use single phase 240V power. MIAQ proposes to include “directly” and “120V” into (2) as a modification to ensure that products powered by external transformer are excluded. These products have been inadvertently included in DOE’s proposed definition. Products without a fan that do not consume energy should be clearly exempt. Electrostatic products use distinctly different technologies and cannot be tested using certain particulates. Commercial products, those delivering more than 400 CFM, the ENERGY STAR limit, are also proposed to be excluded. These products are all distinct from the products covered by ENERGY STAR, which provide air cleaning for human health, rather than for primarily equipment protection. The ENERGY STAR program recognizes the necessary exclusion of Incidental Air Cleaning Products by excluding “combination products” from the program.⁸

In-duct Air Filters are subject to existing performance rating standards and state regulations

Air filters remove particulates from the air stream in ducted forced-air heating or cooling systems in residential and commercial buildings. The removal of particulates, such as dust, from the air stream protects heating, ventilation, and air-conditioning (HVAC) systems from degradation. Air filters ensure the proper operation of these HVAC systems by keeping internal components clean and free of particulate build up that lowers system efficiency, reduces reliability, and diminishes heat transfer. The efficacy of air filters is based on particulate removal. Design of the product is also focused on this efficacy.

Air filters use either mechanical filtration or electrostatic filtration to remove particulates from the air. Mechanical or surface media filtration is the capture of particulates through a dense fiber medium. Typically, the air filter media are pleated, allowing more surface area to capture debris. Electrostatic filtration uses electrostatic precipitation to remove particulates. Electrostatic precipitators charge particles and pull them out of the air stream. Electronic air filters are tested using AHRI Standard 680 because they are incompatible with the conductive loading dust used in ASHRAE 52.2.⁹

There are two elements of air filter performance: (1) the effectiveness of particulate removal from the air and (2) resistance to airflow (or pressure drop) across the filter. One measurement of an air filter’s effectiveness at removing particles from the air is particle size efficiency (PSE), which is the fraction or percentage of particles captured on an air filter. Particle size efficiency is measured across three particle-size bins: 0.3 to 1.0 micrometer (µm), 1.0 to 3.0 µm, and 3.0 to 10.0 µm, as directed by either the ASHRAE 52.2 Standard or the two performance rating standards for residential and commercial/industrial air filter

⁸ EPA defines Combination Product as “An air cleaner that provides an additional function, not related to air purification, within the same housing, such as a humidifier or dehumidifier.” Version 2.0 of ENERGY STAR program requirements can be accessed at https://www.energystar.gov/sites/default/files/ENERGY%20STAR%20Version%202.0%20Final%20Room%20Air%20Cleaners%20Program%20Requirements_0.pdf.

⁹ ASHRAE 52.2-2017 *Method of Testing General Ventilation Air Cleaning Devices for Removal Efficiency by Particle Size*, available at https://www.ashrae.org/File%20Library/Technical%20Resources/COVID-19/52_2_2017_COVID-19_20200401.pdf.

equipment, AHRI Standard 680¹⁰ and AHRI Standard 850,¹¹ respectively. Both AHRI standards establish definitions, classifications, test requirements, rating requirements, minimum data requirements for Published Ratings, including marking and nameplate data, and conformance conditions.

AHRI Standard 680 applies to factory-made Air Filter Equipment and Air Filter Media, as used in such equipment, for removing particulate matter, when used in environmental conditioning of inhabited spaces in residential facilities. The standard evaluates the “combined” performance of Air Filter Equipment in all aspects: Initial Resistance, Final Resistance, Particle Size Efficiency, and Dust Holding Capacity. This offers both the user and specifier a complete view of the Air Filter Equipment for comparison purposes.

AHRI Standard 850 applies to factory-made Air Filter Equipment and Air Filter Media as used in such equipment, for removing particulate matter, when used in environmental conditioning of inhabited spaces in commercial and industrial facilities.

California regulates air filters. Effective on July 1, 2014, the California Building Energy Efficiency Standards began requiring the installation of air filters, in newly constructed residential buildings and certain alterations to residential HVAC systems, that were marked by the manufacturer to indicate particulate filtration efficiency and pressure drop. Effective for air filters manufactured on or after July 1, 2016, the California Appliance Efficiency Regulations began requiring all residential air filters sold or offered for sale in California to be certified to the California Energy Commission (CEC) and marked by the manufacturer to indicate particulate filtration efficiency and pressure drop. The appliance efficiency regulations were intended to harmonize with and complement the building energy efficiency standards. However, in the fall of 2016, the CEC learned that manufacturers of air filters were unable to comply. The regulations were meant to provide manufacturers flexibility when conducting the required tests; however, manufacturers lacked information on how to select which of their air filters were to be tested pursuant to the requirement to test a small, medium, and large air filter. In spring 2017, CEC issued an emergency order to rescind the rule. A new rulemaking docket was opened in 2020 intending to correct regulations for testing, marking, and certification requirements in the appliance efficiency regulations.¹² The CEC’s emergency regulations provide additional time for CEC to collaborate with stakeholders to study, address, and resolve manufacturers’ concerns with the existing testing requirements and allow manufacturers to continue to supply air filters to the California market. While the collaboration between CEC and industry was successful, the rulemaking setting has not concluded.

Since September 2010, California Air Resource Board (CARB) has regulated the amount of ozone produced from indoor air cleaning devices, with the goal to protect public health. All air cleaner models marketed or sold in California, including air cleaners sold via the Internet, after October 18, 2010, must be tested and certified as required by the regulation. In October 2020, the regulation was expanded to include in-duct air cleaning devices, meaning this type of air cleaner must be CARB certified prior to sale

¹⁰ AHRI Standard 680 (I-P/2017): *Performance Rating of Residential Air Filter Equipment*. Available for free download at https://ahrinet.org/App_Content/ahri/files/STANDARDS/AHRI/AHRI_Standard_680_IP_2017.pdf.

¹¹ AHRI Standard 850 (I-P/2013): *Performance Rating of Commercial and Industrial Air Filter Equipment*. Available for free download at https://ahrinet.org/App_Content/ahri/files/STANDARDS/ANSI/ANSI_AHRI_Standard_850_I-P_2013.pdf.

¹² Information on CEC Docket 20-AAER-02 is available, here: <https://www.energy.ca.gov/rules-and-regulations/appliance-efficiency-regulations-title-20/appliance-efficiency-proceedings-0>

to California residents or businesses. There is a 24-month phase-in period for meeting this new requirement, which will end on October 1, 2022.¹³

Manufacturers of air cleaners must already optimize product design for efficacy, compliance with industry performance standards, and to meet existing state regulations. DOE needs to account for potential conflicts caused by multiple regulations, including issues resulting from optimizing products to meet competing goals. Energy efficiency is important, but the main purpose of these products – air cleaning – needs to remain the primary focus for product design.

Request for Comment Issues:

Test Methods

In addition to ANSI/AHAM AC-1-2020, and the four procedures DOE noted in the RFI, AHRI is aware of a few other test methods and regulatory schemes for air cleaners.

Taiwan regulates air cleaners for residential or similar use, defined by CNS 16098 using the metric Clean Air Supply Rate (CASR).¹⁴ The following products are out of scope: (1) those rated/tested CASR is less than 0.26 or more than 12.80; (2) those where the principle of air purification function is either pure ozone, ultraviolet rays, or photocatalyst; and (3) those where the input voltage of the USB port body is DC 5V and does not have a power adapter. CASR is the result calculated by the following steps: Firstly, subtract the particulate matter concentration decay rate by natural phenomena from the total particulate matter concentration decay rate obtained from the test according to this standard to get a remainder. Then multiply the remainder by the internal volume of the test chamber to get the CASR. The calculation result is in response to the amount of clean air per unit time for specific particulate matter and particle size range provided by the operation of the air cleaner and is expressed in m³/min(cmm). CASR is an evaluation of the overall effect of the operation of the tested air cleaner on the decay of the particulate matter concentration in the test chamber. Standby power is also accounted for in the procedure. The regulation goes into force on January 1, 2023. Energy labeling is required to be on the product itself or included in manuals.

Thailand has had a voluntary program¹⁵ in effect since 2020 for 'Air cleaner which are defined in TIS3061-2563 (air cleaner in residential used or related with voltage not exceeding 250V), the Clean Air Delivery Rate (CADR) is tested per TIS 3061:2563 AIR-CLEANER APPLIANCES: PM2.5 REDUCTION EFFICIENCY. Labeling is required for participating products, reporting efficiency in m³/h/W.

MIAQ has also identified several other test methods that may be applicable, including: AHAM AC-3; AHAM AC-5-2021; AHAM AC-4; GB/T18801-2015 (Chinese); NRCC-54013 (Canadian); ISO 16000-36; ISO/CD 16000-43; ISO/CD 16000-44; NF-B44-200:2016; NF EN 16846-1:2017; JEM 1467 2015 (Japan); IEC 63086-2 (gases); SPS-KACA002 2016 Korean; ISO/TC 142 - IEC 63086; ASHRAE 52.2; ASHRAE 52.2 with

¹³ Information regarding CARB Air Cleaners regulation is available at <https://ww2.arb.ca.gov/our-work/programs/air-cleaners-ozone-products/air-cleaner-regulation-ab-2276>.

¹⁴ CNS 16098: *Air Cleaners for household and similar use – Methods for measuring the performance*, available at https://www.cnsonline.com.tw/?node=result&typeof=common&locale=zh_TW

¹⁵ <http://labelno5.egat.co.th/new58/wp-content/uploads/update/product/airpure.pdf>

optional Appendix J; ASHRAE 52.2 proposed Appendix; ISO 16890; AHRI Standard 850; AHRI Standard 680/681 – 2017; ASHRAE 145.2; ISO 10121; and ASHRAE 185.1.

Technology Assessment and Efficiency Analysis

MIAQ cannot provide concrete information on the availability or lack thereof of technologies for improving energy efficiency of air cleaners for non-portable products until the scope and definition are appropriately tailored to exclude products inappropriate for regulation.

It is also difficult for MIAQ to provide data and information regarding annual energy use estimates for air cleaners, particularly for products not covered by the ENERGY STAR program, such as non-portable products (wall mounted, ceiling-mounted, and whole home units), while products are inappropriately included in the scope. For example, Electrostatic Air Cleaners installed in ducts and without a built-in Fan may consume about 20W in operation. They only operate when the HVAC fan is on for heating, cooling, or ventilating.

Ducted products, with fans primarily used for ventilating, cooling, and heating, employ different technologies than portable products, with distinctly different energy use patterns. The energy calculations would need to account for different hours of operations for different products, specific to each product. For example, an oven hood may operate for 20 minutes a day and be used to remove grease particulates while a portable air cleaner might be set to run continuously.

Madison Indoor Air Quality appreciates the opportunity to provide these comments. If you have any questions regarding this submission, please do not hesitate to contact me.

Sincerely,



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